Technische Hochschule Ingolstadt

LOMA
Learning and Optimization in Manufacturing

AImotion Bavaria

Assignment 11

Software Development

Task 1 - Debugging

A strategy for detecting logic errors (Task 1a):

- Define pieces of your program you want to test for logic errors
- Define pairs of inputs and expected outputs for these pieces
- Compare the expected and the actual outputs

- Print the current state of your calculations at various points during execution
- Use the debugger in your IDE

Debugging logic errors – Task 1b)

```python
def vector_addition(vectorA, vectorB):
    result = [None] * len(vectorA)

    for i in range(1, len(vectorA)):
        result[i] = vectorA[i] + vectorB[i]

    return result
```

- **Range for the indices starts at 1**
- List of fixed length is defined
- Iteration over the list is done using indices
- Problems if lists have different length
- Results are written directly to the list instead of using append()

Debugging logic errors – Task 1b)

```
def vector_addition(vectorA, vectorB):
    result = list()

    if len(vectorA) != len(vectorB):
        print("Vectors must have the same length")
        return None

    for vectorA_i, vectorB_i in zip(vectorA, vectorB):
        result.append(vectorA_i + vectorB_i)

    return result
```

- List is now dynamic
- Same vector size is enforced
- Direct iteration over both lists simultaneously using zip
- Result is written to the list using append

→ Works correctly, is easier and more robust!

Debugging logic errors – Task 1c)

```
def calculate_overall_price(area, price_per_sqm=40, tax=0.15):
    net_price = area * price_per_sqm
    gross_price = net_price * tax
    return gross_price

area = 500
price_per_sqm = 85
tax = 0.10

overall_price = calculate_overall_price(area, price_per_sqm)
print(f"The overall price is {overall_price}")
```

- Gross price is only the amount of tax
- tax is not passed to the function → may lead to wrong results

Debugging logic errors – Task 1c)

```
def calculate_overall_price(area,price_per_sqm=40, tax=0.15):
    net_price = area * price_per_sqm
    gross_price = (net_price * tax) + net_price
    return gross_price

area = 500
price_per_sqm = 85
tax = 0.10

overall_price = calculate_overall_price(area, price_per_sqm, tax)
print(f"The overall price is {overall_price}")
```

- Gross price is calculated correctly
- tax is actually passed to the function